CrossMaps: A Confidence-Aware Open-Vocabulary Semantic Mapping for Rover Navigation

by Jan-Niklas Klein
Robotic Sensing and Knowledge Transfer Learning in Safety-Critical Environments - WiSe 2025/26

Motivation

- How can our rover get a **semantic understanding** of the environment?
- **Classic SLAM** (*Simultaneous Localization and Mapping*):
  - Geometry-only
- **Classic Object Detection** / Segmentation:
  - Limited to predefined object classes
- How can the robot build a **robust, language-conditioned semantic map while moving**?
  - Adapting a **VLMaps**-style semantic navigation
  - **Open-vocabulary semantic maps** with natural language indexing, building on top of preexisting *RTAB-Map* SLAM
    - Building a **semantic memory**

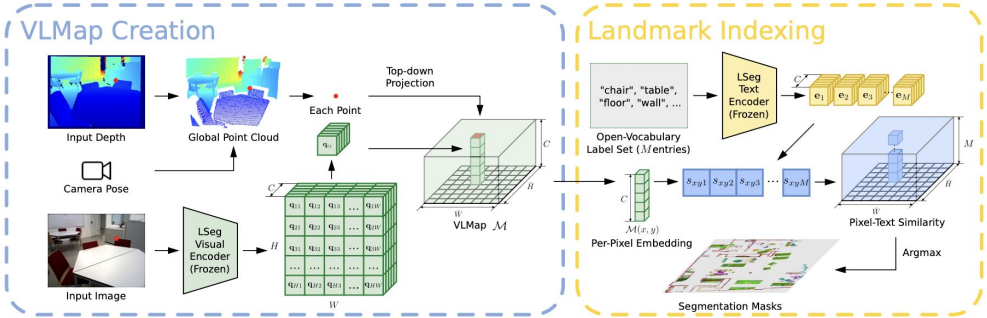
Background

VLMaps and related works

Visual Language Maps – VLMaps

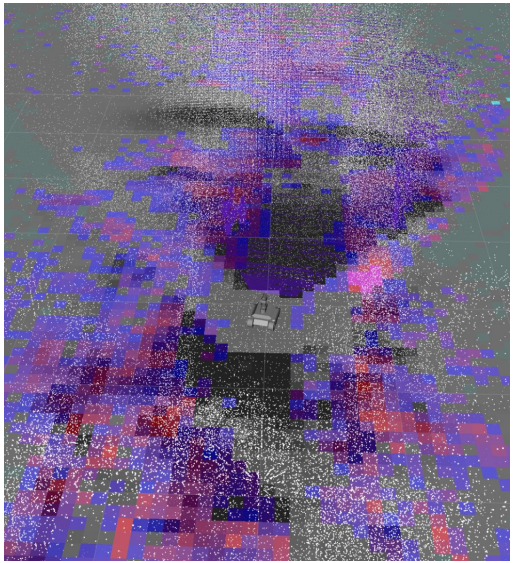
Background

- Paper by Google scientists from 2023 (Huang et al.)
- Transformer models can **encode knowledge into embedding space**
- ***Core idea:*** Add **semantic embeddings** to every spatial point
  - Use pre-trained **visual transformer** model out-of-the-box
  - Use RGB-D camera to **back-project embedded pixels** into 3D space
  - Built a **language-queryable 3D map** / grid for navigation



Challenges

- How do we make the system **robust against noise**?
- How can we make it **online**, updating while driving?
- How can we make it **stable while moving**?
- How do we keep it adaptable, when we have **dynamic objects**?
- What is a good tradeoff between **plasticity and robustness**?
- How do we deal with **drift and loop closures**?



Related papers and methods

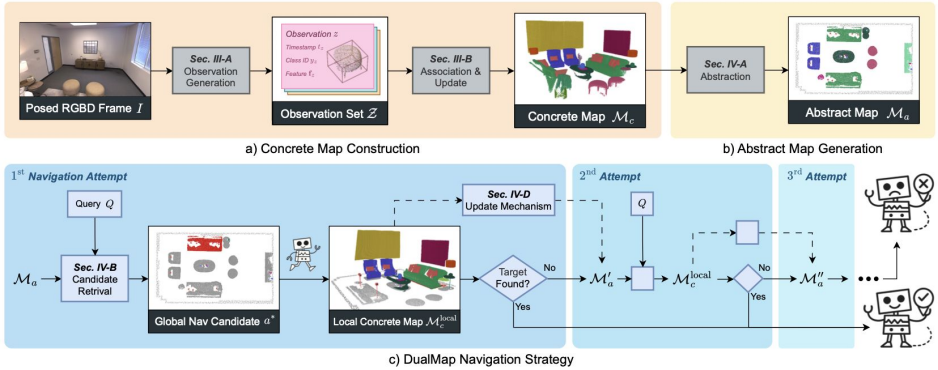
Background

❖ Several papers extend the ideas introduced in VLMaps:

Paper	Online	Loop Closure	Geometry	Environment Assumption	Confidence-aware	Memory Model	Multi-Agent
VLMaps	✔ partial	✖	Occupancy grid + CLIP map	Static	✖	✖	✖
OpenFusion	✔	✖	TSDF	Static	✔	✖	✖
OpenScene	✖ offline	✖	Point cloud / voxel CLIP	Static	✖	✖	✖
LERF	✖	✖	NeRF	Static	✖	✖	✖
OVO (2025)	✔	✔	TSDF + SLAM	Partial	✔	✖	✖
FindAnything (2025)	✔	✔	Object-centric map	Partial	Partial	✖	✖
DualMap (2025)	✔	✔	Dynamic semantic map	Dynamic	✔	Partial	✖

DualMap

- Jiang et al. (2025)
- ***Core idea:*** Create a fine-grained **concrete** and an **abstract semantic map**
  - Combines **open-vocabulary**, **online** and **dynamic navigation**
  - Abstract map derived from concrete map:
    - **Abstract Map**: Global structural cues are sufficient for high-level planning
    - **Concrete Map**: Accurate object details retrieved locally through online perception
    - **Intra-object stability checks**



Concept

Short Term & Long Term Memory

Concept

STM → LTM

- **Detailed local online semantic map**
  - **Confidence** + multi-viewpoint tracking
- **Dynamic objects** can appear / disappear
  - Slightly delayed to reduce noise
- **Time-decay** on confidence to filter noise
  - Confidence in observations decay over time
- Relatively **noisy**

→

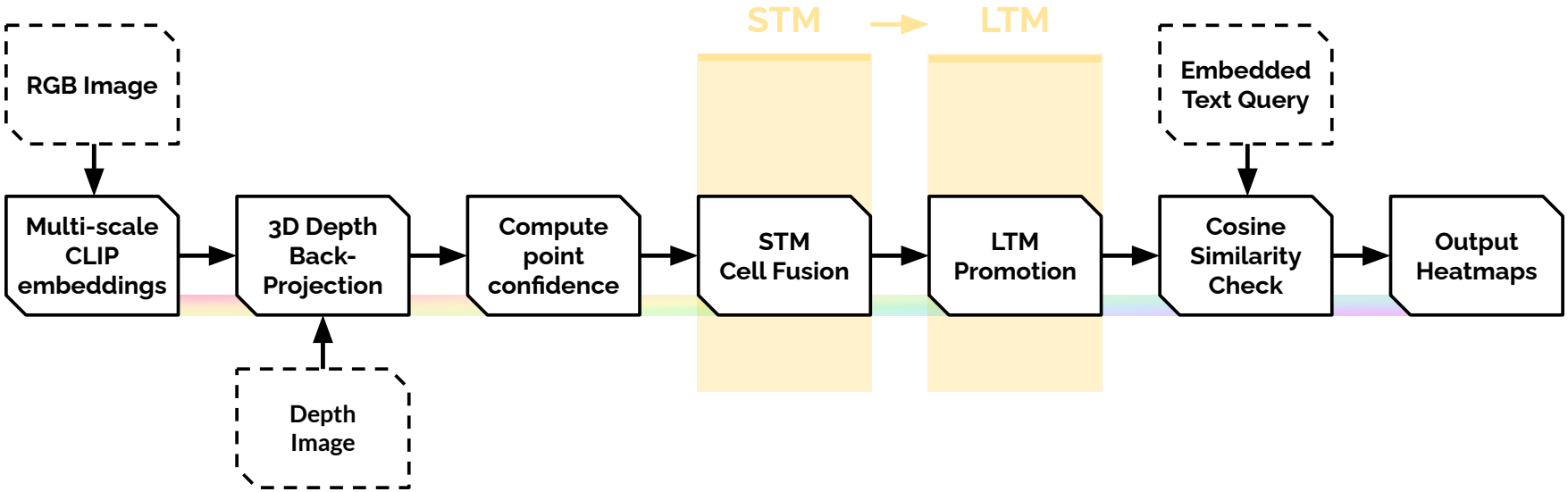
- **Robust** and **confident long term map**
- More sparse and **slower but reliable**
- **Higher-level semantic landmarks** for navigation
  - Easily shareable for multi-agent scenarios
  - Time-stamps: last seen
- **No time-decay, stable**
- **Noise-free**

Demo Video

Algorithm + Conceptual Details

Pipeline Overview

Algorithm



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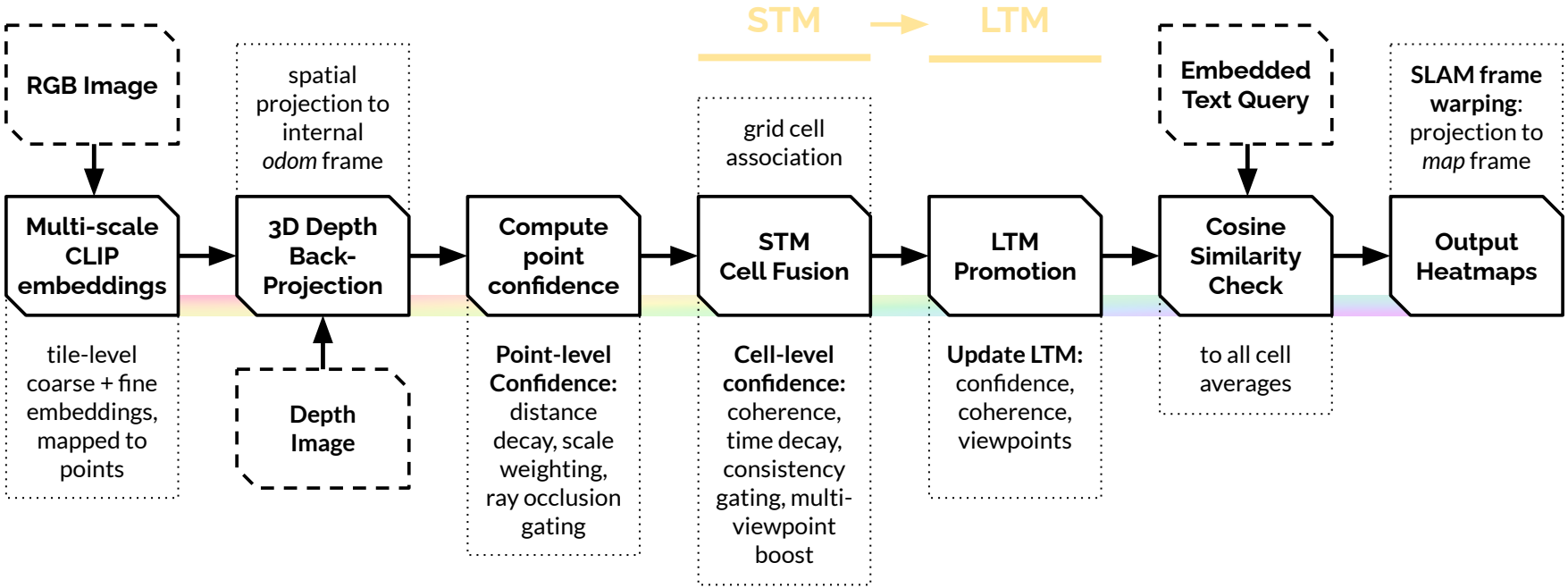
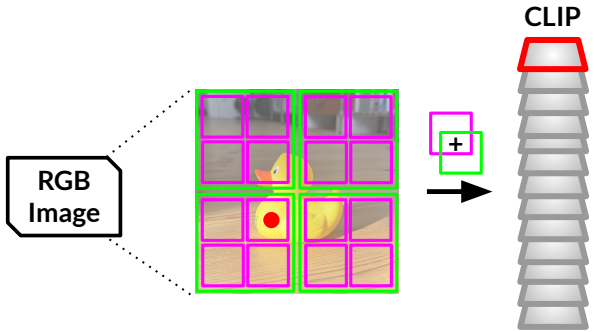


Visual Transformer Model + CLIP

- Method can use **off-the-shelf vision language models**
  - Visual transformer models
  - Model: OpenAI ViT-B-32 *(Inference RAM: ~3.8GB, VRAM: ~1.5GB)*
  - *Better but slower:* OpenAI ViT-L-14
- Embeds into **CLIP feature space** (**CLIP embeddings**)
- Input: Sequences of **RGB-D images**
  - VLMaps: LSeg pixel-level embeddings on RGB image (more precise)
  - My approach: **Tile-level embeddings** (**Coarse + Fine tile embeddings**)
    - Faster computation + more robust for rover movement
  - Then **backproject to 3D space** with depth image and camera pose

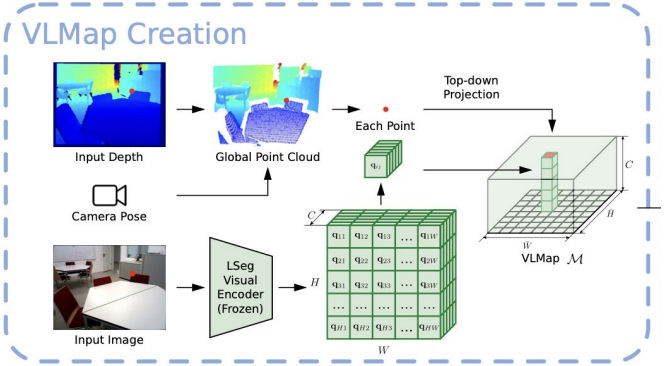


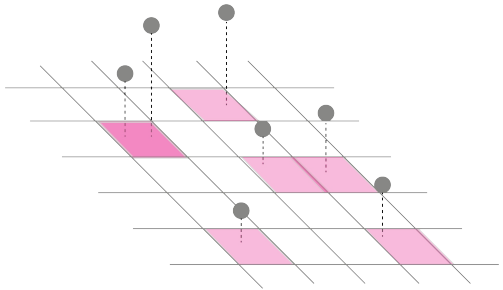
$$\mathbf{e}_{mix} = w_c\,\hat{\mathbf{e}}_c + (1 - w_c)\,\hat{\mathbf{e}}_f$$

$$\hat{\mathbf{e}} = \frac{\mathbf{e}_{\text{mix}}}{\|\mathbf{e}_{\text{mix}}\|_2 + \varepsilon}$$

3D space: Find grid cell

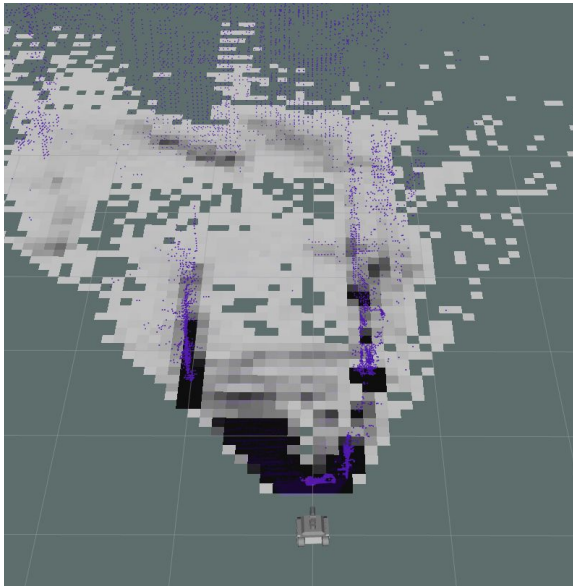
- **2D top-down projection to grid cells** for limiting resource consumption and aggregated computations
- **Static occupancy grid** of size H x W x C
  - Could be extended to dynamic hash grid
- Step 1: Transform **local depth point cloud** from depth image into **world frame** (using camera pose)
- Step 2: **Top-down project to 2D occupancy grid**





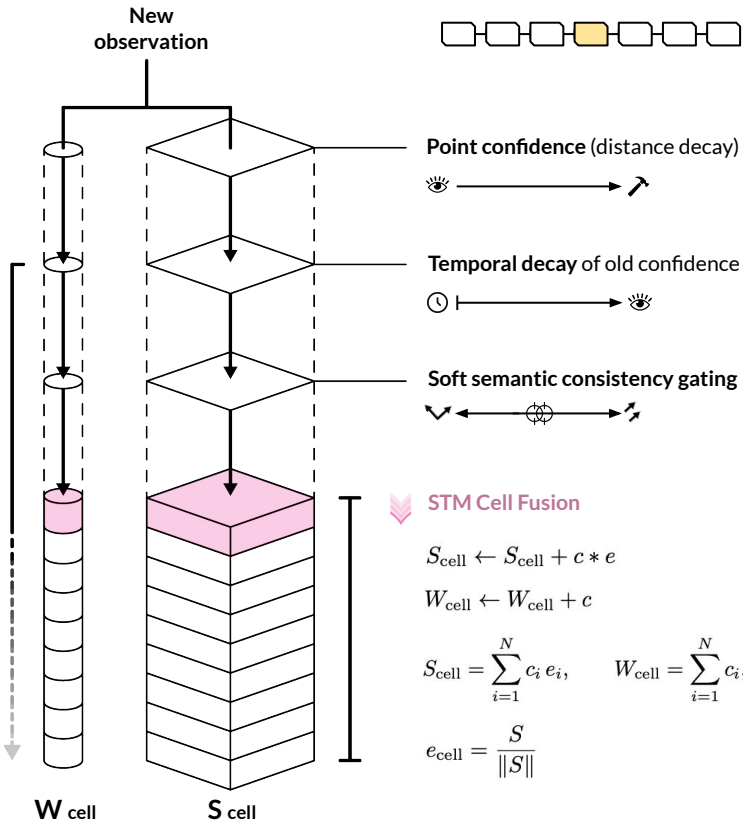
Factors of Confidence

- Direct **confidence measures** and **stability checks**
- **Geometric + Semantic + Temporal Confidence**
- **Geometric Confidence**
  - Distance decay, ray occlusion gating, multi-viewpoint boost
- **Semantic Confidence**
  - Semantic consistency gating / weighting (*agreement before fusion*), Coherence estimation (*agreement after fusion*)
- **Temporal Confidence**
  - Time-based confidence decay in STM
  - Long-term memory without decay



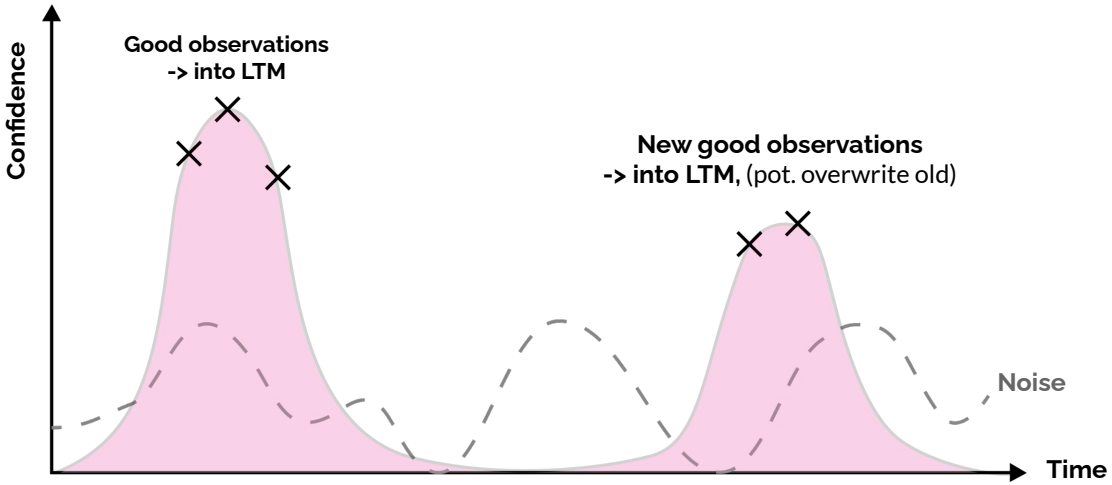
STM semantic cell fusion

- **Weighted average embeddings** for multiple points on same grid cell
  - **Confidence-weighted update**
    - (c -> confidence, e -> embedding)
  - **Before fusion**:
    - Point confidence (distance decay)
    - Temporal decay of old cell confidence
    - Soft consistency gating (on cosine similarity)
  - **Weighted fusion** (Aggregated sums)
  - **After fusion**:
    - Viewpoint boost (from bitmap of viewing angles)
    - Coherence estimation (do observations agree?)



LTM cell promotion: Confidence

- **Promotion from STM cell to LTM cell**, if
  - **Confidence** + **coherence** sufficient and **multiple viewpoint** observation
  - STM:

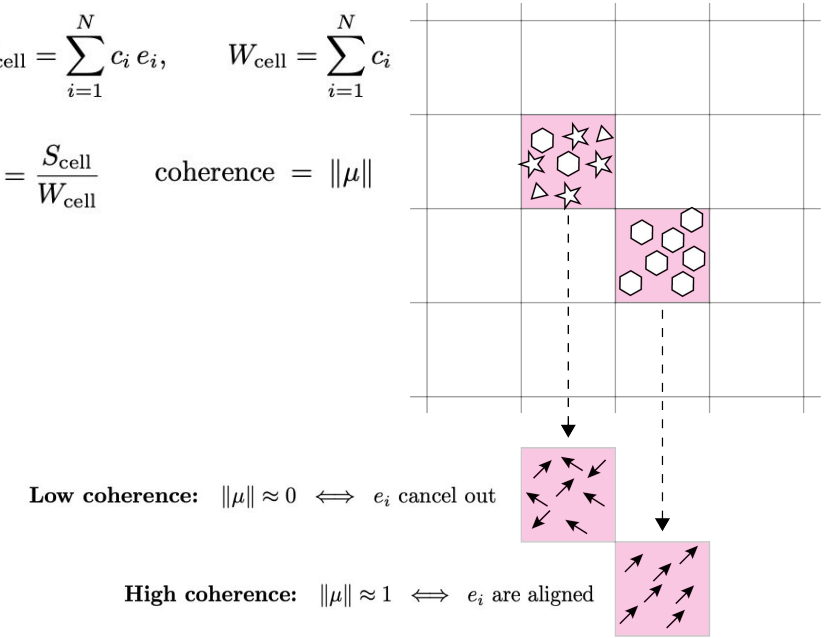


LTM cell promotion: Coherence

- Measuring **semantic consistency** within a cell
  - How aligned are the observations?
    - Agreement in embedding space
    - **Angular variance estimation**
    - Norm of cell mean
- Downweight noisy cells, improve reliability
- With **time decay**, coherence **shifts towards recent observations**

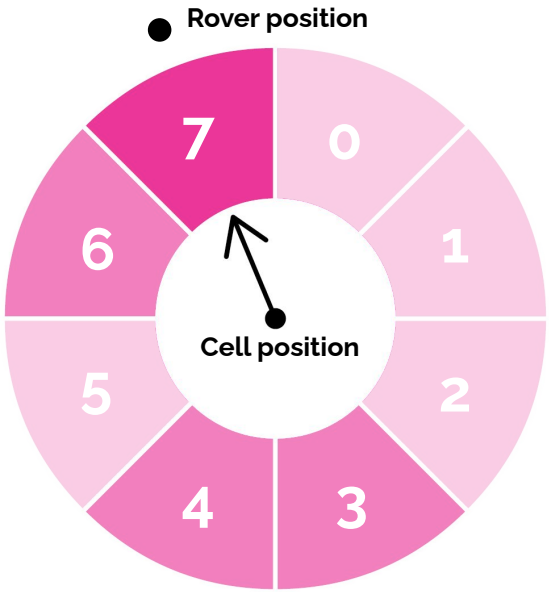
$$S_{\text{cell}} = \sum_{i=1}^{N} c_i \, e_i, \qquad W_{\text{cell}} = \sum_{i=1}^{N} c_i$$

$$\mu = \frac{S_{\text{cell}}}{W_{\text{cell}}} \qquad \text{coherence} = \|\mu\|$$



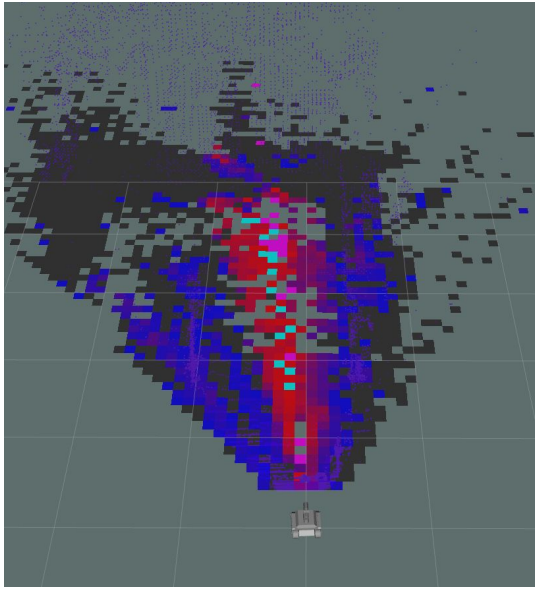
LTM cell promotion: Viewpoint bins

- Convert rover view angle to cell viewpoint bin
- Save in bitmask per cell, eg. `00011011`
  - Track viewpoint angles
  - Use for confidence boost in STM
  - Promotion check for LTM



Querying and Embedding Similarity

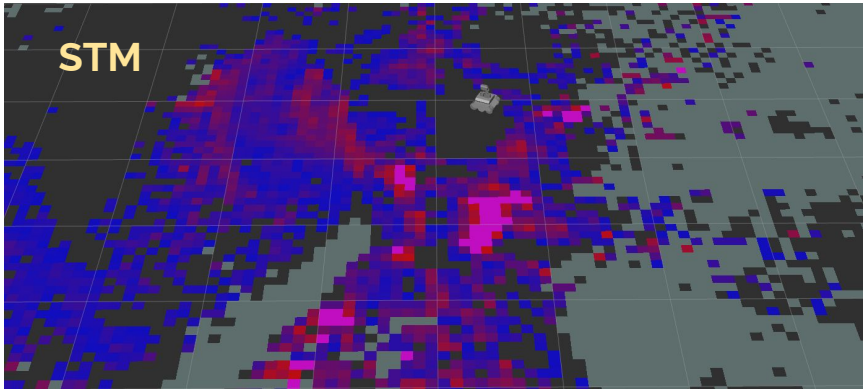
- Convert text into embeddings with **CLIP text encoder**
  - OpenAI CLIP ViT-B/32 text encoder
- Create **semantic similarity heatmaps** over grid
  - Compute **cosine similarity** between **query text** and **each cell**
    - $s = \cos\left(\hat{e}_{\text{cell}}, e_{\text{text}}\right)$
  - For **heatmap**, **coherence** factor is used (agreement)
    - Keep interesting but low-confident areas $s \leftarrow s * \text{coherence}$
  - For **top1 cell**, **confidence** is used directly:
    - rank $r_i = s_i * c_i$
  - **Natural language spatial indexing**

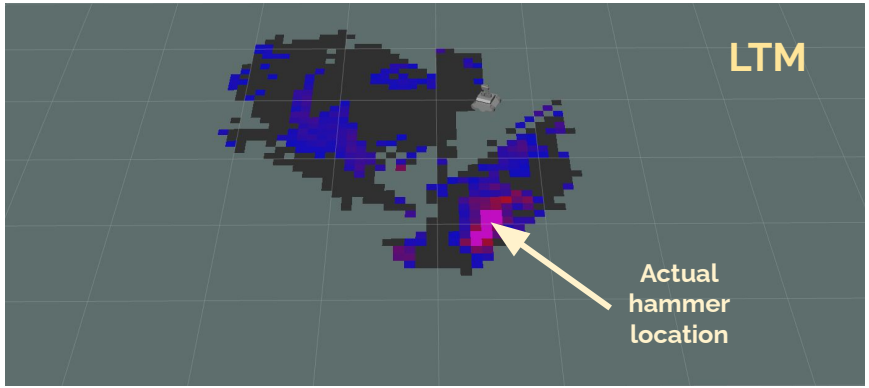


STM vs LTM Heatmap

Detailed, noisy STM heatmap vs. **Sparse, reliable LTM heatmap**

Query: “hammer”



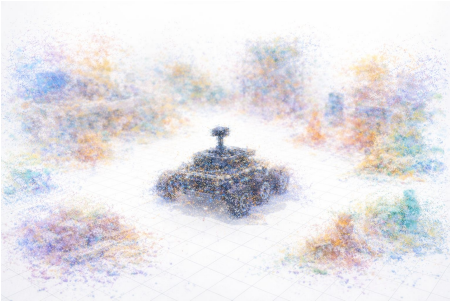


Local areas of interest

Higher-level navigation
Potential multi-rover map sharing

Frame handling & SLAM correction

- **Pipeline**: Camera -> odom -> semantic grid -> map
  - Everything is internally in odom, then gets transformed and published to map
- **Loop closures**
  - While driving, small drift accumulates in odom
    - Odom: Stable internal / local frame for accumulating data
    - Map: Globally corrected target frame from SLAM
  - **On loop closure: warp odom / semantic grid to map** using *RTAB-Map TF Tree*
    - RTAB-Map does the loop closures for us



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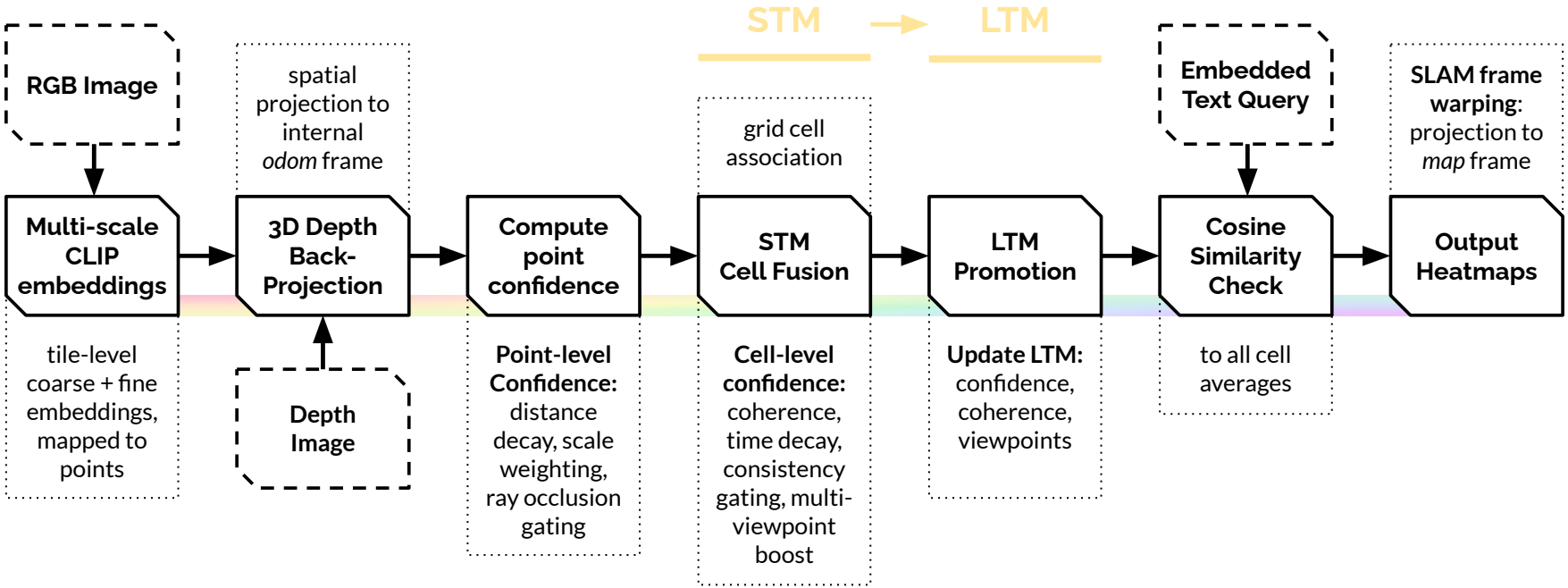

Future work

- Extend to **multi-agent shared maps** with loop closures
  - **Long term memory is shared** / merged between rovers
- Experiment with extending **STM time decay** from confidence to **semantics** as well
  - Potentially better **dynamic object handling**
- Reconsider **replacing tiles with segmentation**: Use **Segment Anything Model** as pre-step in pipeline, running the ViT only on the masks
- Potentially **multiple semantical hypotheses per cell**, ranked by confidence
  - Eg.: **Negative hypothesis** could dominate in case an object is removed / not there anymore
  - Potentially better handling of ambiguity and object disappearance
- Implement **Nav2** integration

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